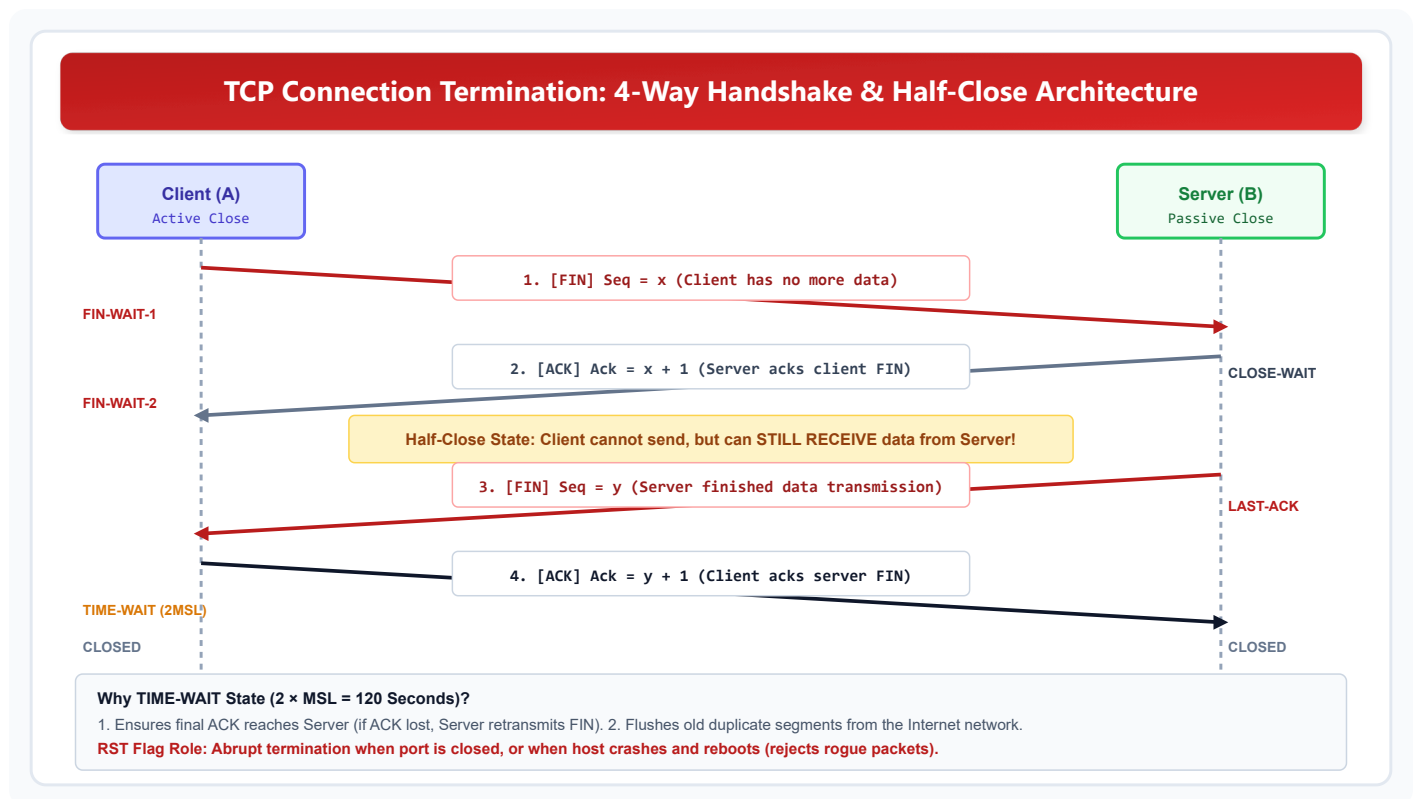


Module 04: TCP Connection Termination and Half-Close

Unit 4: Transport Layer

Termination & State Dynamics

Four-Way Handshake with FIN & ACK, Half-Close Architecture, TIME-WAIT (2MSL) State Engineering, and Reset (RST) Flag Usage.



1. 4-Way Handshake Steps

Step	Direction	Flag	Equations	States
1	Client -> Server	FIN = 1	Seq = x	Client: FIN-WAIT-1
2	Server -> Client	ACK = 1	Ack = x + 1	Client: FIN-WAIT-2, Server: CLOSE-WAIT

3	Server -> Client	FIN = 1	Seq = y	Server: LAST-ACK
4	Client -> Server	ACK = 1	Ack = y + 1	Server: CLOSED, Client: TIME-WAIT (2MSL)

2. TIME-WAIT (2MSL) Rationale

Reason 1: Allows retransmitting final ACK if Server's FIN times out.

Reason 2: Prevents old duplicate segments from corrupting future connections.

RST Flag: Abrupt teardown when connecting to closed port or during crash recovery.